

Speaker Notes

AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media

Bachelor's diploma defence, 2026

Slide 1. AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media

Purpose:

Open the defence, name the topic, and immediately set the correct research scope.

What to say:

Good morning. The topic of our diploma work is AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media. The work studies how machine learning can support directional prediction for thermoelastic wave behavior in simplified geological media. I would like to emphasize the scope from the beginning: this is a research prototype and a comparative framework. It is not presented as a site-calibrated simulator. The focus is on physical interpretation, model comparison, and estimating directional characteristics under simplified assumptions.

Key points:

- The work connects thermoelastic physics, geological parameters, and neural surrogate models.
- The main target is directional prediction between a source and a probe point.
- The presentation treats the system as a research prototype.

Transition to next slide:

I will start by explaining why this topic is relevant.

Slide 2. Relevance of the Topic

Purpose:

Explain why thermoelastic wave prediction in rocks is worth studying.

What to say:

Thermally affected rocks appear in several applied contexts: geothermal energy, underground construction, thermal stimulation, and geophysical interpretation. When temperature changes inside a rock, it can influence stress, deformation, and wave behavior. A direct coupled numerical simulation is useful, but it can be expensive when many materials, source positions, or observation directions must be compared. This motivates an AI-guided approach, where neural models estimate selected directional responses and help compare scenarios faster. The important point is that the AI model supports the analysis; it does not replace the physical problem formulation.

Key points:

- Temperature changes can create mechanical effects in geological media.
- Repeated coupled simulations are expensive for comparative studies.
- The work uses AI-guided prediction as a supporting research tool.

Transition to next slide:

This relevance leads to the main research problem.

Slide 3. Research Problem

Purpose:

State the central question and narrow it to a defensible bachelor thesis scope.

What to say:

The central question is how to estimate directional characteristics of thermoelastic wave propagation without rerunning a full coupled numerical simulation for every new scenario. Real geological media are complex: they may be porous, fractured, anisotropic, and saturated with fluids. Modelling all these effects is outside the scope of this thesis. Therefore, the work uses a simplified two-dimensional effective-medium formulation. Within that controlled setting, the prototype compares several neural surrogate approaches for directional prediction.

Key points:

- The problem is framed as directional prediction, not full field simulation.
- Real rocks are more complex than the simplified model.

- The simplified setup makes controlled model comparison possible.

Transition to next slide:

Next, I will formulate the aim and the tasks of the work.

Slide 4. Aim and Tasks

Purpose:

Show what the diploma work tries to achieve and how the study is organized.

What to say:

The aim is to develop and justify an AI-guided framework for directional prediction of thermoelastic wave propagation in geological media. To reach this aim, the work first establishes the physical and geological basis of the problem. Then it formulates a simplified two-dimensional prediction task. After that, it uses COMSOL-derived synthetic data for controlled comparison. Finally, it compares PINN, FNO, MeshGraphNet, and Transformer-style surrogate routes and interprets their results and limitations.

Key points:

- The aim combines physical reasoning with model comparison.
- The task is built around a simplified 2D formulation.
- The models are compared under one shared prediction contract.

Transition to next slide:

Before the theory, I will define the object and subject of the study.

Slide 5. Object and Subject of the Study

Purpose:

Clarify what is being studied and what specific aspect the thesis focuses on.

What to say:

The object of the study is thermoelastic wave propagation in geological media represented by effective rock parameters. The subject is narrower: AI-guided directional prediction of thermal and displacement response along a selected source-probe path. This distinction is important. We do not model a full natural formation with all geological details. We study a simplified comparative framework where the material is represented by effective parameters and the prediction is made for a defined direction.

Key points:

- Object: thermoelastic wave behavior in geological media.
- Subject: directional prediction along a source-probe path.
- The work is intentionally limited to a simplified comparative framework.

Transition to next slide:

Now I will explain the basic physical logic of thermoelastic coupling.

Slide 6. Thermoelastic Coupling Logic

Purpose:

Introduce the physical chain from heating to wave-like mechanical response.

What to say:

This slide shows the basic logic of thermoelastic coupling. A local thermal source changes the temperature field. Temperature change produces thermal expansion, and if the expansion is constrained by the surrounding material, stress appears. Stress gradients then create mechanical displacement, and this response propagates through the medium. The formula at the bottom is not meant to be read as a full equation. It summarizes the chain: temperature leads to thermal strain, thermal strain contributes to stress, and stress drives displacement.

Key points:

- Heating changes the temperature field.
- Thermal expansion can produce stress.

- Stress gradients drive displacement and propagation.

Transition to next slide:

The next group of slides explains the physical and geological parameters used in this chain.

Slide 7. Geological Media and Effective Parameters

Purpose:

Show that geological materials enter the prototype through effective scalar parameters.

What to say:

Here we show a compact example of the geological material parameters used in the study. Density, P-wave velocity, thermal conductivity, and heat capacity differ between rock types such as granite, basalt, sandstone, and limestone. In the prototype, these are effective scalar parameters. They allow us to compare material behavior in a controlled way. However, they should not be interpreted as a complete field calibration for a real geological site.

Key points:

- Rock types differ in density, stiffness, wave speed, and thermal properties.
- The parameters are effective values for comparison.
- The table supports modelling assumptions rather than field calibration.

Transition to next slide:

The next slide expands this idea into the full material table.

Slide 8. Full Geological Media Parameter Table

Purpose:

Document the broader material catalogue used for the comparative framework.

What to say:

This table gives the extended catalogue of effective geological media parameters. It includes igneous, sedimentary, and metamorphic rocks, together with mechanical and thermal quantities such as density, elastic moduli, Lamé parameters, thermal expansion, conductivity, heat capacity, and wave velocities. The key idea is not to discuss every number in detail. The table shows that the model inputs are physically meaningful and material-dependent. Missing values are shown explicitly, which is important for transparency.

Key points:

- The table links rock groups with mechanical and thermal parameters.
- Elastic constants are derived under isotropic assumptions.
- Missing thermal data are marked rather than hidden.

Transition to next slide:

After listing the parameters, I will explain how they control propagation.

Slide 9. Rock Properties Controlling Propagation

Purpose:

Explain the physical role of mechanical and thermal rock properties.

What to say:

Rock properties influence propagation through both mechanical and thermal mechanisms. Density controls inertia: a denser medium resists acceleration more strongly. Elastic stiffness controls how strongly the rock resists deformation. Porosity, cracks, and saturation can reduce effective stiffness and change wave behavior. On the thermal side, conductivity controls heat transfer, heat capacity controls thermal storage, and thermal expansion controls how temperature change becomes strain. In real rocks, bedding, foliation, aligned pores, fractures, and stress state can also create directional effects.

Key points:

- Mechanical properties control inertia and stiffness.
- Thermal properties control heat spreading and thermal strain.

- Geological structure can create directional behavior.

Transition to next slide:

To connect this with wave propagation, I will briefly discuss P-waves and S-waves.

Slide 10. Elastic Waves in Geological Media

Purpose:

Introduce compressional and shear waves using simple physical interpretation.

What to say:

Elastic wave propagation is commonly described using P-waves and S-waves. P-waves are compressional waves, where particle motion is mainly parallel to the propagation direction. S-waves are shear waves, where particle motion is transverse and depends on shear rigidity. The formulas show that wave velocity depends on stiffness and density. Physically, a stiffer rock usually transmits disturbances faster, while porosity and compliant cracks often reduce effective velocity. These relations also justify using wave speed as a reference for travel-time sanity checks.

Key points:

- P-waves involve compression and volume change.
- S-waves depend on shear rigidity.
- Wave velocities depend on elastic moduli and density.

Transition to next slide:

Next, I will add the thermal part of the coupled problem.

Slide 11. Heat Conduction and Thermal Expansion

Purpose:

Explain how heat transfer enters the thermoelastic formulation.

What to say:

The thermal part is described by heat conduction. Fourier's law says that heat flux is directed opposite to the temperature gradient and depends on thermal conductivity. The heat equation describes how temperature evolves in time under conduction and a heat source. The mechanical connection appears through thermal expansion. We do not need to read the equations term by term. Their physical meaning is that heat diffuses through the rock, and a temperature increase can produce thermal strain. If the surrounding medium restricts this strain, thermal stress appears.

Key points:

- Thermal conductivity controls the heat flux.
- The heat equation describes temperature evolution.
- Thermal expansion links temperature change to strain.

Transition to next slide:

Now I will summarize the key physical parameters in one place.

Slide 12. Key Physical Parameters

Purpose:

Make the parameter roles easy to remember for the committee.

What to say:

This slide summarizes the main parameters. Thermal diffusivity combines conductivity, density, and heat capacity. It describes how quickly a temperature disturbance spreads through the medium. Density also controls inertia in the mechanical response. Bulk and shear moduli control compressional and shear stiffness. Thermal expansion controls how strongly a temperature change produces strain. In the prediction framework, these parameters become input features, so the neural models receive physically interpretable information rather than only abstract numerical labels.

Key points:

- Thermal diffusivity describes the rate of heat spreading.
- Density and stiffness influence mechanical wave behavior.
- Thermal expansion is the bridge from heat to deformation.

Transition to next slide:

The next slide combines these ideas in the thermoelastic equations.

Slide 13. Thermoelastic Coupling

Purpose:

Present the theoretical backbone without overloading the speech with derivation.

What to say:

This slide gives the key thermoelastic relations used as theoretical background. The temperature perturbation is measured relative to the initial temperature. The stress relation contains the usual elastic part, controlled by Lamé parameters, and a thermal stress term, controlled by the coupling coefficient. The equation of motion then links stress gradients to acceleration and displacement. In simple terms, the equations say that temperature changes can modify stress, and stress drives motion. These relations motivate why a physics-informed baseline is useful in the comparison.

Key points:

- Stress contains both elastic and thermal contributions.
- The thermal coupling coefficient is related to bulk modulus and expansion.
- The equation of motion connects stress gradients with displacement.

Transition to next slide:

I will now define the mathematical scope used in the thesis.

Slide 14. Mathematical Scope and Assumptions

Purpose:

Make the simplifications explicit and defensible.

What to say:

The model is formulated as a two-dimensional continuum problem. The displacement field is in-plane, so the model tracks horizontal and vertical displacement components. Small deformation and linear isotropic thermoelasticity are used as first-order approximations. This is a deliberate simplification. The model does not explicitly resolve pores, fractures, fluids, plasticity, damage, or full anisotropic constitutive behavior. These assumptions make the problem tractable for a bachelor-level comparative study and allow the model outputs to be interpreted consistently.

Key points:

- The thesis uses a 2D continuum formulation.
- Small deformation and linear isotropic thermoelasticity are assumed.
- Complex geological effects are acknowledged as limitations.

Transition to next slide:

With the assumptions defined, I will list the primary fields used in the model.

Slide 15. Primary Fields in the 2D Model

Purpose:

Connect the theoretical variables with the prediction outputs.

What to say:

This table lists the main fields of the simplified model. Temperature describes the thermal state of the rock, and temperature perturbation drives thermal strain and stress. The displacement components describe the mechanical response at the probe point. Strain and stress are internal physical quantities derived from the displacement field and material law. Finally, the direction variables describe the source-probe geometry: distance, unit direction, and azimuth. These fields connect the theory from the first chapters with the prediction

methodology.

Key points:

- Temperature and displacement are the main observable response fields.
- Strain and stress provide the physical link inside the model.
- Direction variables define what is meant by directional prediction.

Transition to next slide:

Now I will explain the directional prediction task more directly.

Slide 16. Directional Prediction

Purpose:

Define the source-probe query and what the prototype estimates.

What to say:

Directional prediction is built around a source point and a probe point. The vector from the source to the probe defines the direction of interest. From this vector, we compute the distance, unit direction, and azimuth angle. The prototype then estimates the thermal and displacement response associated with that query. The important point is that we are not predicting every possible field value in a complete geological volume. We are estimating directional characteristics for a specified material, geometry, source, probe, and observation time.

Key points:

- A source and a probe define the directional query.
- Distance and azimuth are explicit geometric quantities.
- The predicted response is interpreted along that direction.

Transition to next slide:

Next, I will show how this physical idea becomes a machine learning prediction task.

Slide 17. From Physics to a Prediction Task

Purpose:

Translate the physical problem into an input-output mapping for surrogate models.

What to say:

The prediction task is written as a mapping from input features to estimated outputs. The input includes material parameters, source location, probe location, propagation direction, and observation time. The output includes estimated temperature, displacement, directional response, and travel time. This formulation is intentionally narrower than solving the complete coupled PDE system at all field points. It is designed for structured comparison of predictive estimates under the same input-output contract.

Key points:

- Inputs combine material, geometry, direction, and time.
- Outputs include temperature, displacement, and travel-time estimates.
- The task is predictive and comparative, not a full physical solver.

Transition to next slide:

To train and compare the models, the work uses a controlled synthetic dataset.

Slide 18. COMSOL-Based Synthetic Dataset

Purpose:

Explain where the controlled training and comparison data come from.

What to say:

The dataset is derived from a controlled COMSOL thermoelastic experiment. The domain is a two-dimensional plane-strain rock sample with size one meter by one meter. A heated internal source generates a time-dependent thermal and mechanical response. For each node and time step, the exported outputs include temperature, displacement, velocity, stress, strain, and material features. The purpose of this dataset is to provide a common reference for comparing the surrogate models under the same conditions.

Key points:

- The dataset comes from a controlled COMSOL-based setup.
- The model uses a 2D plane-strain domain.
- The data provide a shared basis for comparison.

Transition to next slide:

After data generation, the exports must be represented in a model-ready form.

Slide 19. Data Representation and Preprocessing

Purpose:

Show how raw simulation exports are prepared for different model families.

What to say:

The COMSOL CSV exports are aligned into a single node set. Dynamic features are assembled into a spatio-temporal tensor, while static material and geometry features are stored separately. Before training, the feature channels are normalized using the mean and standard deviation. This step is important because the variables have different physical units and scales. The same processed dataset can then support several model types: coordinate-based models, grid-based models, and graph-based models.

Key points:

- Raw exports are aligned by nodes and time steps.
- Dynamic and static features are represented separately.
- Normalization makes model training and comparison more stable.

Transition to next slide:

Now I will describe the comparative prototype framework.

Slide 20. Comparative Prototype Framework

Purpose:

Explain that all models are evaluated under one shared workflow.

What to say:

The prototype uses one workflow for all model routes. The frontend, backend orchestrator, material catalog, and model services follow a shared input-output contract. This is important for fair comparison: the same source-probe geometry, the same material presets, and the same normalized response format are used for every route. The goal is not to show a production platform, but to create a consistent research environment for comparing AI-guided directional prediction methods.

Key points:

- All model routes receive comparable inputs.
- Outputs are normalized into the same response structure.
- The framework supports fair model comparison.

Transition to next slide:

Next, I will introduce the four model families compared in the work.

Slide 21. Compared Model Families

Purpose:

Give a high-level comparison of the model roles without going into implementation details.

What to say:

The work compares four model families. PINN is used as the physics-informed baseline because its objective includes physical residual terms. FNO is a supervised operator model for structured grid representations and global field patterns. MeshGraphNet represents the mesh as a graph and uses message passing to model local interactions. The Transformer-style route treats geometry and state variables as query features and is useful for fast flexible inference. These are compared as surrogate routes, not as equivalent physical solvers.

Key points:

- PINN provides the physics-informed baseline.
- FNO, MeshGraphNet, and Transformer are surrogate routes for comparison.
- The comparison focuses on role and behavior, not deep architecture detail.

Transition to next slide:

I will now explain the PINN route in more detail because it is the physics-informed baseline.

Slide 22. PINN as the Physics-Informed Baseline

Purpose:

Explain why PINN is treated differently from the purely supervised surrogate routes.

What to say:

The PINN route uses a point-wise neural network that takes coordinates, time, and material parameters as input and predicts temperature and displacement components. Its training objective combines supervised loss with residual terms related to velocity, elasticity, and thermal behavior. This should not be interpreted as an exact PDE solution. The careful interpretation is that PINN is a physics-informed baseline: it includes physical structure in the loss and therefore provides a useful reference for the other surrogate approaches.

Key points:

- PINN maps coordinates, time, and material parameters to response fields.
- The loss includes supervised and physics-related residual terms.
- PINN is a physics-informed baseline, not a complete solver.

Transition to next slide:

The next slide summarizes the other surrogate routes.

Slide 23. Other Surrogate Routes

Purpose:

Describe FNO, MeshGraphNet, and Transformer-style models at the right level for defence.

What to say:

The other routes are treated as neural surrogate approaches. FNO is designed for field-to-field operator learning on structured two-dimensional representations. MeshGraphNet is useful when a mesh can be represented as a graph, so that local node interactions are modeled through message passing. The Transformer-style route uses coordinate queries and feature tokens, which makes it flexible and fast for inference. In this thesis, these models are compared as alternative predictive routes. They are not described as physically equivalent numerical solvers.

Key points:

- FNO targets structured field patterns.
- MeshGraphNet targets mesh-aware local interactions.
- Transformer-style prediction provides fast query-based inference.

Transition to next slide:

Now I will move from methodology to the experimental setup and results.

Slide 24. Experiment Status

Purpose:

Summarize the evaluation setup and make caveats transparent before showing plots.

What to say:

The evaluation uses forty scenarios per material and four materials: granite, sandstone, limestone, and basalt. With four model routes, this gives six hundred forty model responses under the same contract. The important technical note is that every plotted response is an active model output, not a mock fallback. The operating envelope is still limited to the two-dimensional plane-strain setup. Also, FNO is included in the comparison, but its output scale is currently miscalibrated, so it should be interpreted as a diagnostic branch rather than a reliable final route.

Key points:

- The comparison covers $40 \times 4 \times 4 = 640$ model responses.
- All plotted responses are active outputs.
- FNO is included, but its current calibration is limited.

Transition to next slide:

The first result concerns predicted travel times.

Slide 25. Travel-Time Predictions

Purpose:

Explain what the travel-time plot shows and what conclusion should be drawn.

What to say:

This plot compares predicted travel time by model and material. The main observation is that the differences between model families are larger than the difference between basalt and sandstone in the current prototype. PINN and Transformer produce much smaller travel-time estimates, while FNO and MeshGraphNet produce larger values. This does not mean one model is universally correct. It shows that, under the current setup, model formulation strongly influences directional travel-time estimates. Therefore, model comparison is necessary before drawing physical conclusions.

Key points:

- The plot compares travel-time estimates across models and materials.
- Model-family differences dominate the material difference here.
- The result motivates careful interpretation and validation.

Transition to next slide:

To judge accuracy, the next slide compares the estimates with an analytical reference.

Slide 26. Accuracy Against Analytical Reference

Purpose:

Present the key quantitative result and its cautious interpretation.

What to say:

Here the predicted travel times are compared with a simple analytical reference, $t_{gt} = r/V_P$. This reference is not a full thermoelastic validation; it is a useful sanity check based on distance and P-wave velocity. In this comparison, PINN has the lowest median relative error, around two percent, and remains within the ten percent band across the balanced sweep. Transformer is faster but less accurate for this metric. FNO and MeshGraphNet show larger errors in the current configuration. The result suggests that the physics-informed baseline is the most consistent route for this specific travel-time comparison.

Key points:

- The reference is a sanity check, not complete validation.
- PINN shows the best agreement with the analytical reference.
- Other routes need further calibration or time-conditioning.

Transition to next slide:

Accuracy is only one side of the comparison; inference time is also important.

Slide 27. Inference Latency and Time Behavior

Purpose:

Compare practical response time and explain the fixed-horizon issue.

What to say:

This plot shows inference latency by model. All routes return predictions in milliseconds on the development machine, so the prototype is suitable for interactive comparison. The Transformer route is the fastest, followed by MeshGraphNet, FNO, and PINN. However, latency is not the only criterion. PINN is also the only route that changes smoothly with requested observation time. The other routes currently behave more like fixed-horizon

predictors. This means they may be fast, but their temporal behavior needs additional development for time-dependent directional studies.

Key points:

- All routes are fast enough for interactive prototype use.
- Transformer has the lowest median latency.
- PINN currently has the most meaningful time-conditioned behavior.

Transition to next slide:

Before concluding, I will state the main threats to validity.

Slide 28. Threats to Validity

Purpose:

Show scientific caution and define the limits of the results.

What to say:

There are several threats to validity. First, the training data are limited and come from a pilot COMSOL-derived configuration, not from a complete calibrated geological database. Second, the model uses two-dimensional plane-strain assumptions and effective material parameters. Third, the current FNO route has a normalization issue and is not yet physically reliable. Fourth, the analytical reference based on r/V_P is only a sanity check, not a full validation of thermoelastic behavior. These limitations are important because the results should be interpreted within the simplified assumptions of the study.

Key points:

- The dataset is limited and synthetic.
- The physical formulation is simplified to 2D effective media.
- The analytical reference is useful but limited.

Transition to next slide:

With these limitations in mind, I can summarize the conclusions.

Slide 29. Conclusions

Purpose:

State the main scientific outcomes of the work.

What to say:

The thesis develops a physically grounded comparative framework for AI-guided directional prediction of thermoelastic wave propagation. Within the tested assumptions, PINN is the strongest physics-informed baseline and shows the best agreement with the analytical travel-time reference. Transformer is the fastest route and is useful for rapid query-based inference. MeshGraphNet remains a useful mesh-aware surrogate candidate, but needs further time-conditioning. FNO should be treated as a diagnostic branch until its output normalization is corrected. Overall, the work demonstrates a careful prototype for comparing neural surrogate approaches rather than a final geophysical simulator.

Key points:

- The main contribution is a comparative AI-guided framework.
- PINN performs best against the travel-time reference.
- Other models remain useful candidates with specific limitations.

Transition to next slide:

Finally, I will outline the most important next steps.

Slide 30. Future Work

Purpose:

Show how the prototype can be developed after the diploma.

What to say:

The first direction is to retrain and recalibrate the two-dimensional FNO route, because its current output scale is not reliable. The second direction is to add stronger time-conditioned rollout behavior to MeshGraphNet. The third direction is to extend the material catalogue with more complete thermoelastic parameters and validation data. It would also be useful to compare the prototype against additional analytical and numerical baselines. The final message is that the prototype estimates directional characteristics of thermoelastic wave propagation within simplified assumptions, and it provides a basis for further research.

Key points:

- Recalibrate FNO and improve time behavior in MeshGraphNet.
- Extend material data and validation sources.
- Add stronger analytical or numerical baselines.

Transition to next slide:

This completes the presentation, and I am ready for questions.

Slide 31. Thank You

Purpose:

Close the presentation and invite questions.

What to say:

Thank you for your attention. I will be happy to answer your questions about the physical assumptions, the model comparison, the dataset, or the interpretation of the results.

Key points:

- Close politely.
- Signal readiness to discuss assumptions and limitations.
- Keep answers consistent with the research prototype framing.

Transition to next slide:

Questions from the committee.

Possible questions from the committee

1. Why did you choose directional prediction instead of full-field simulation?

Because the thesis focuses on a narrower and more defensible research task. Full-field thermoelastic simulation is computationally and physically more complex. Directional prediction allows us to compare source-probe responses, travel time, and model behavior under controlled assumptions.

2. Is the prototype validated on real field data?

No. The current work uses COMSOL-derived synthetic data and analytical sanity checks. The results should be interpreted within simplified modelling assumptions. Field data would be an important next step, but it is outside the current scope.

3. Why is PINN called a physics-informed baseline?

PINN is called a physics-informed baseline because its training objective includes residual terms connected with thermoelastic relations. This means that physical structure is included in the learning objective, but the result is still interpreted as a baseline model inside the simplified prototype.

4. Why does FNO perform poorly in the current results?

The current two-dimensional FNO route has a calibration and output-scale issue. Therefore, it is kept in the comparison as a diagnostic branch. Its architecture is still relevant, but this implementation needs retraining and normalization correction.

5. What is the role of the analytical reference $t=r/V_P$?

It is a simple travel-time sanity check based on distance and P-wave velocity. It is not a complete thermoelastic validation, but it helps identify whether predicted travel times are physically plausible in a basic isotropic approximation.

6. What are the main limitations of the work?

The main limitations are the simplified 2D setting, effective isotropic material parameters, limited synthetic data, fixed-horizon behavior in some surrogate routes, and the absence of direct calibration against real geological measurements.

7. What is the main contribution of the diploma work?

The main contribution is a physically grounded comparative framework for AI-guided directional prediction. It combines thermoelastic theory, geological material parameters, a controlled dataset, and several neural surrogate routes under one consistent comparison workflow.

Short version of the speech

Good morning. The topic of our diploma work is AI-Guided Prediction of Thermoelastic Wave Propagation in Geological Media. The work presents a research prototype for AI-guided directional prediction. It is not a site-calibrated simulator. The goal is to estimate selected directional characteristics of thermoelastic wave propagation under simplified assumptions and to compare several neural surrogate approaches.

The physical motivation is that temperature changes in rocks can create thermal expansion, stress, displacement, and wave-like mechanical response. This is relevant for geothermal systems, underground engineering, and geophysical interpretation. A full coupled numerical simulation can be expensive when many materials and source-probe directions must be compared. Therefore, the thesis formulates a narrower prediction task.

The model uses effective geological material parameters such as density, elastic moduli, thermal conductivity, heat capacity, thermal expansion, and P-wave and S-wave velocities. These parameters describe how different rock types respond mechanically and thermally. The simplified formulation is two-dimensional, uses small deformation and linear isotropic thermoelasticity, and does not explicitly resolve pores, fluids, fractures, plasticity, or full anisotropy.

Directional prediction is defined by a source point and a probe point. Their difference gives the direction, distance, and azimuth. The prototype estimates temperature, displacement, travel time, and directional response for that query. The data are based on a controlled COMSOL-derived two-dimensional thermoelastic setup, exported and normalized for several model families.

The comparison includes PINN, FNO, MeshGraphNet, and a Transformer-style route. PINN is used as the physics-informed baseline because its objective includes thermoelastic residual terms. FNO, MeshGraphNet, and Transformer-style models are treated as surrogate routes for comparison, not as physically equivalent numerical solvers.

The experiments use forty scenarios per material, four materials, and four model routes, giving six hundred forty model responses. In the travel-time comparison, model-family differences are larger than the material difference in the current prototype. Against the simple analytical reference $t=r/V_P$, PINN shows the best agreement and remains within the ten percent band in the balanced sweep. Transformer is the fastest route, while MeshGraphNet and FNO need additional development for time behavior and calibration.

The main limitations are the limited synthetic dataset, the simplified 2D assumptions, the current FNO normalization issue, and the fact that the analytical reference is only a sanity check. The conclusion is that the work provides a physically grounded comparative framework for AI-guided directional prediction within simplified assumptions. Future work should recalibrate FNO, improve time-conditioned MeshGraphNet behavior, extend material data, and compare against additional analytical or numerical baselines.